

Project #3

Enterprise Windows Infrastructure Administration Lab

Objective

Design, administer, maintain, troubleshoot, and document a Windows Server infrastructure that simulates the day-to-day responsibilities of a Junior Windows System Administrator.

This project focuses on **administration**, not initial deployment.

Technologies

Windows Server 2022

Active Directory

Group Policy

DNS

DHCP

VMware Workstation Pro

PowerShell

Windows 11

pfSense

Documentation Structure

Project 03 – Enterprise Windows Infrastructure Administration Lab

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— Documentation

—	Screenshots
—	PowerShell Scripts
—	Resources
—	ProjectReport.pdf
—	README.md

Documentation Files

01-Project-Design.odt
02-Infrastructure-Overview.odt
03-Server-Roles.odt
04-PowerShell-Administration.odt
05-System-Administration-Tasks.odt
06-Troubleshooting.odt
07-Lessons-Learned.odt

Administration Modules

Instead of "Help Desk Tickets," we'll organize this project into **administration modules**.

Module 1

Active Directory Administration

Examples:

- User lifecycle
 - Security groups
 - OU delegation
 - Fine-Grained Password Policy
 - Restore deleted objects
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Module 2

Group Policy Administration

- Administrative Templates
 - Loopback Processing
 - WMI Filters
 - Security Filtering
 - GPO Backup
 - GPO Restore
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Module 3

DNS Administration

- Forward Lookup Zones
 - Reverse Lookup Zones
 - Conditional Forwarders
 - Aging & Scavenging
 - Troubleshooting
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Module 4

DHCP Administration

- Reservations
 - Exclusions
 - Scope Options
 - Lease Management
 - Troubleshooting
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Module 5

File Server Administration

- DFS Namespace
 - DFS Replication
 - NTFS Permissions
 - Shadow Copies
 - File Screening
 - Disk Quotas
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Module 6

Windows Server Administration

- Windows Services
 - Server Manager
 - Event Viewer
 - Performance Monitor
 - Task Scheduler
 - Windows Server Backup
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Module 7

PowerShell Administration

Examples:

- Bulk user creation
 - Bulk password reset
 - Export AD users
 - Export group members
 - Disable inactive accounts
 - Generate reports
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Module 8

Infrastructure Troubleshooting

Examples:

- DNS failures
 - DHCP failures
 - Group Policy failures
 - Authentication issues
 - File share issues
 - Replication issues
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GitHub Repository

Enterprise-Windows-Infrastructure-Administration-Lab

Skills this project demonstrates

By the end, you'll be able to show experience with:

- Windows Server Administration
 - Active Directory Administration
 - Group Policy Management
 - DNS Administration
 - DHCP Administration
 - PowerShell Administration
 - File Server Administration
 - Windows Infrastructure Troubleshooting
 - VMware Administration
 - Enterprise Documentation
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What comes after Project #3

Once this is complete,

Project #4 – Wazuh SIEM & Endpoint Monitoring Lab